NavFart A1

Problem 1: Attitude control of satellite

1.1 $\dot{q} = T_q(q)\omega$

$$I_g \dot{\omega} - S(I_g \omega)\omega = \tau$$

$$T(q_b^n) = \frac{1}{2}\begin{bmatrix} -\varepsilon^T \\ \eta I_3 + S(\varepsilon) \end{bmatrix}, \quad q = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \tau = 0$$

$$\begin{bmatrix} \dot{\eta} \\ \dot{\varepsilon}_1 \\ \dot{\varepsilon}_2 \\ \dot{\varepsilon}_3 \end{bmatrix}_b^n = T(q_b^n)\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \Rightarrow \dot{\varepsilon} = \frac{1}{2}(\eta I_3 + S(\varepsilon))\omega$$

$$\dot{\omega} = I_g^{-1}(\tau + S(I_g \omega))$$

$q = [1\ 0\ 0\ 0]^T$ and $\tau = 0$:

$$\dot{x} = \begin{bmatrix} \dot{\varepsilon} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}\eta^{(1)}\omega \\ I_g^{-1} S(I_g \omega) \end{bmatrix} = 0$$

$\Rightarrow \omega = 0$

$$x_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \varepsilon \\ \omega \end{bmatrix}_0$$

Non-linear state-space model

$$\dot{x} = \begin{bmatrix} \frac{1}{2}\left[\sqrt{1-\varepsilon^T\varepsilon}\, I_3 + S(\varepsilon)\right]\omega \\ I_g^{-1} S(I_g \omega)\omega + I_g^{-1}\tau \end{bmatrix}$$

Linearization about $x = x_0$:

$$\dot{x} \approx f(x_0, \tau_0) + A(x - x_0) + B(\tau - \tau_0)$$

$$A = \left.\frac{\partial f(x, \tau)}{\partial x}\right|_{x = x_0,\ \tau = \tau_0} \qquad B = \left.\frac{\partial f(x, \tau)}{\partial \tau}\right|_{x = x_0,\ \tau = \tau_0}$$

$$A = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{\partial \dot{\varepsilon}}{\partial \varepsilon} & \frac{\partial \dot{\varepsilon}}{\partial \omega} \\ \frac{\partial \dot{\omega}}{\partial \varepsilon} & \frac{\partial \dot{\omega}}{\partial \omega} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{6}{mL^2} & 0 & 0 \\ 0 & \frac{6}{mL^2} & 0 \\ 0 & 0 & \frac{6}{mL^2} \end{bmatrix} \quad m = 10\,kg,\ L = 0.1\,m$$

$$\frac{6}{mL^2} = 60$$

$$\dot{x} = Ax + B\tau$$

$$\begin{bmatrix} \dot{\varepsilon} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}\omega \\ 60\tau \end{bmatrix}$$

1.2 $\tau = -k_d \omega - k_p \varepsilon$ $\quad k_p > 0,\ k_d = k_d I_3,\ k_d > 0$

$k_p = 5 \cdot 10^{-5},\ k_d = 9 \cdot 10^{-4}$

$\tau = -Kx$ where $K = [k_p I_3 \quad k_d I_3]$

$\dot{x} = Ax + B(-Kx) \Rightarrow \dot{x} = (A - BK)x$

Find poles of $A - BK = \begin{bmatrix} 0_{3\times3} & \frac{1}{2} I_3 \\ -60 k_p I_3 & -60 k_d I_3 \end{bmatrix}$

$= \begin{bmatrix} 0_{3\times3} & 0{,}5\, I_3 \\ -0{,}003\, I_3 & -0{,}054 I_3 \end{bmatrix}$

For each axis

$$\begin{bmatrix} \dot{\varepsilon}_i \\ \dot{\omega}_i \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0{,}5 \\ -0{,}003 & -0{,}054 \end{bmatrix}}_{A_i} \begin{bmatrix} \varepsilon_i \\ \omega_i \end{bmatrix}$$

$\det(\lambda I - A_i) = 0$

$\lambda^2 + 0{,}054\lambda + 0{,}0015 = 0 \Rightarrow \lambda = \dfrac{-0{,}054 \pm \sqrt{0{,}054^2 - 4(0{,}0015)}}{2}$

$\lambda_{1,2} = -0{,}027 \pm 0{,}02777i$

$\lambda_{3,4} =$ —— 11 ——

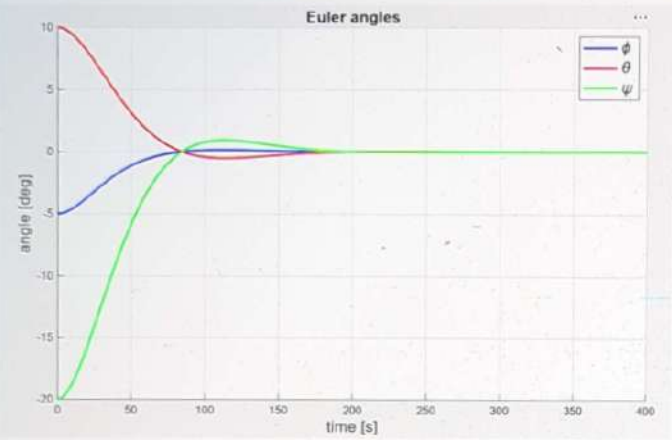
$\lambda_{5,6} =$ —— 11 ——

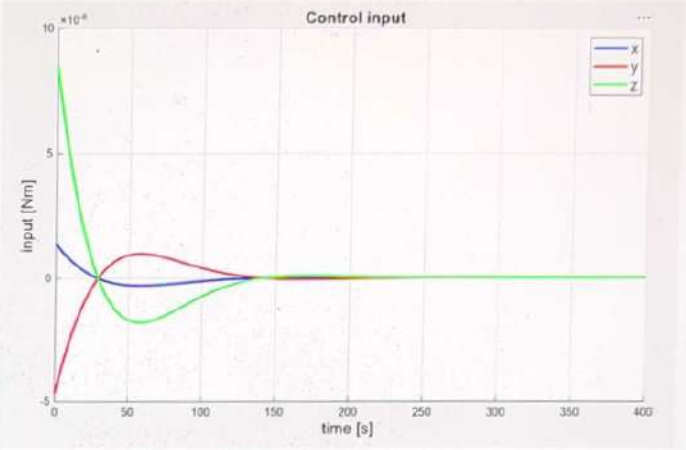
Should do this in matlab

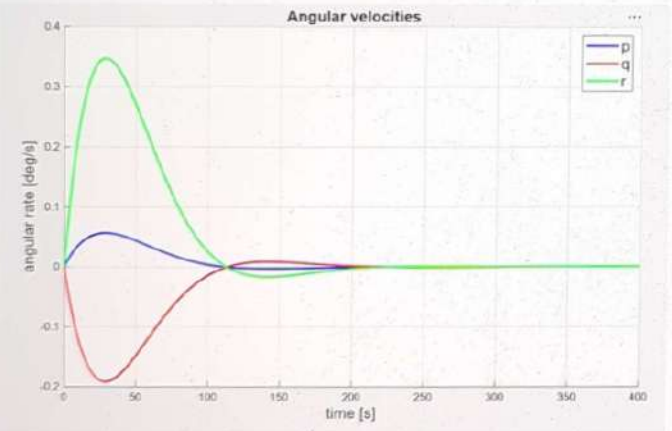
Would prefer real negative poles because that avoids oscillations and overshoot. Complex poles are okay if damping is high

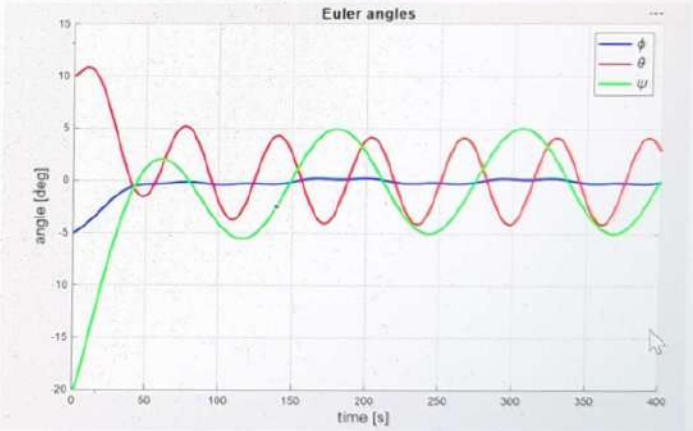
Assignment 1 NavFart

1.3









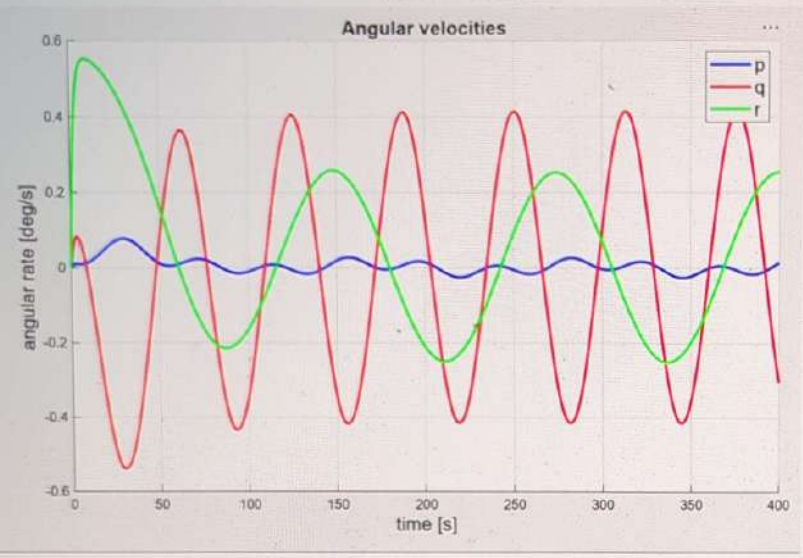
1.3 The behavior mathes my expectations because it converges to the reference with some oscillations.
To follow non-zero reference, you would use error quaternion $\tilde{q} = \bar{q}_d \otimes q$:
$\tau = -K_d \omega - k_p \tilde{\varepsilon}$

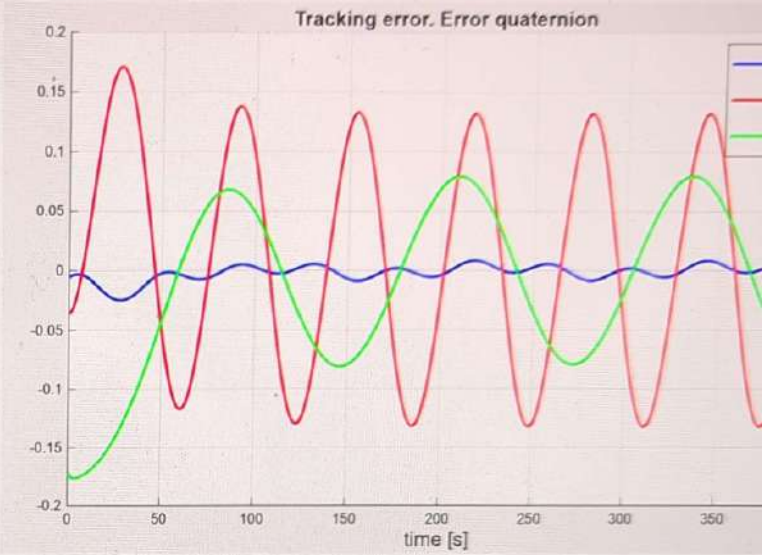
1.4 $\tilde{q} = \bar{q}_d \otimes q = \begin{bmatrix} \eta_d \eta + \varepsilon_d^T \varepsilon \\ \eta_d \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix} + \eta \begin{bmatrix} -\varepsilon_{d1} \\ -\varepsilon_{d2} \\ -\varepsilon_{d3} \end{bmatrix} + \begin{bmatrix} 0 & +\varepsilon_{d3} & -\varepsilon_{d2} \\ -\varepsilon_{d3} & 0 & +\varepsilon_{d1} \\ +\varepsilon_{d2} & -\varepsilon_{d2} & 0 \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix} \end{bmatrix}$

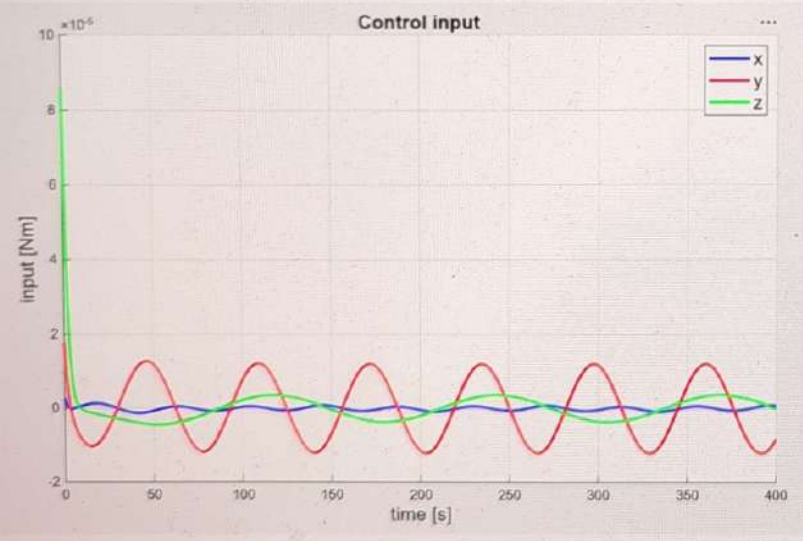
$= \begin{bmatrix} \eta_d \eta + \varepsilon_{d1} \varepsilon_1 + \varepsilon_{d2} \varepsilon_2 + \varepsilon_{d3} \varepsilon_3 \\ \eta_d \varepsilon_1 - \eta \varepsilon_d + \varepsilon_{d3} \varepsilon_2 - \varepsilon_{d2} \varepsilon_3 \\ \eta_d \varepsilon_2 - \eta \varepsilon_{d2} - \varepsilon_{d3} \varepsilon_1 + \varepsilon_{d1} \varepsilon_3 \\ \eta_d \varepsilon_3 - \eta \varepsilon_{d3} + \varepsilon_{d2} \varepsilon_1 - \varepsilon_{d1} \varepsilon_2 \end{bmatrix}$

After convergence $q = q_d$ $\tilde{q} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

1.5


Angular velocities
p
q
r
angular rate [deg/s]
time [s]


Tracking error. Error quaternion
time [s]

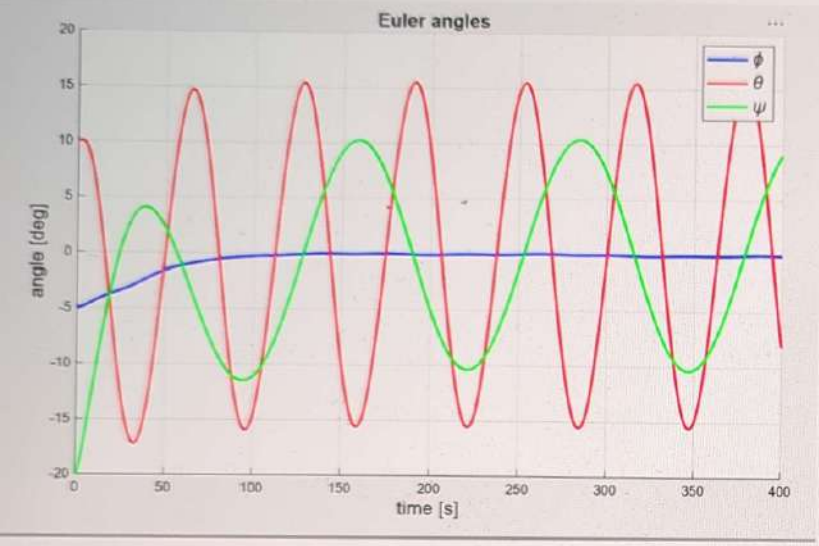

Control input
x
y
z
input [Nm]
time [s]

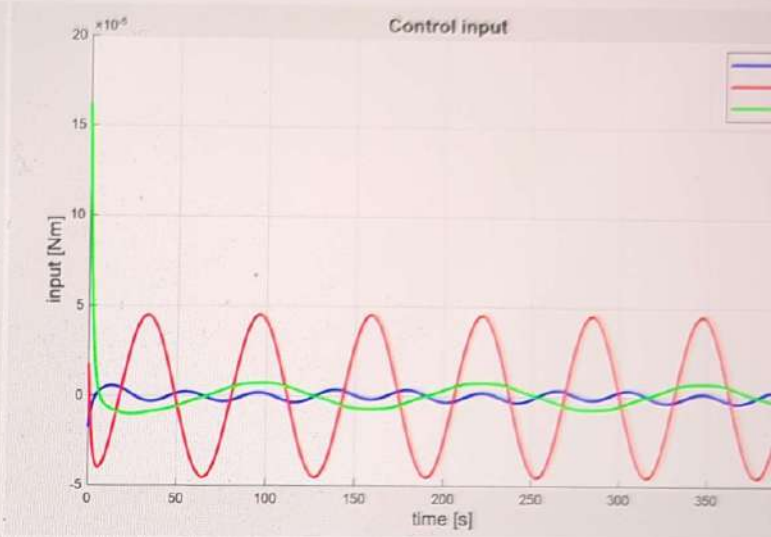
1.5 System is stable, but does not perfectly track the reference. Phase lag and reduced amplitude. Problem is that $K_d \omega$ damps rotation.

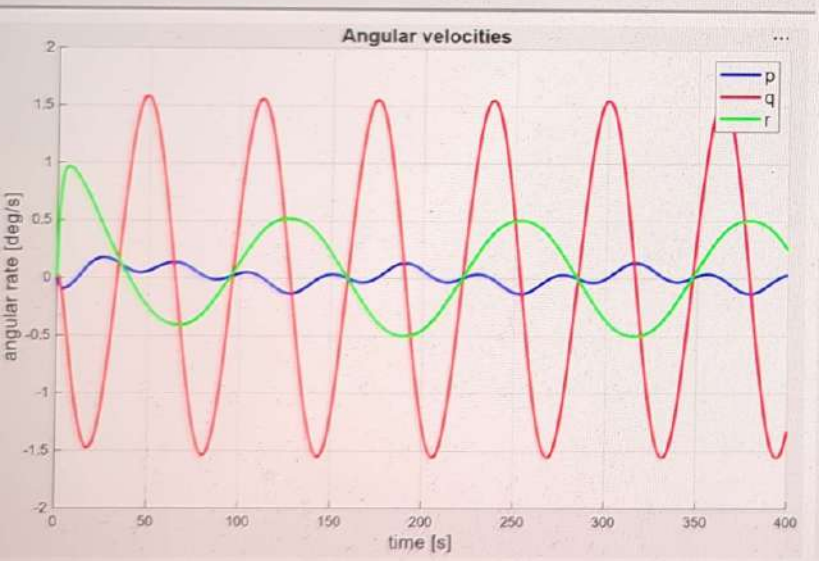
1.6 Modified controller gives improved tracking Euler angles follow their reference trajectory amplitude is correct and tracking error is small. Small periodic tracking error because controller does not account for angular acceleration required by time-v reference. Controller could be improved by adding model based feed forward term including $\dot{\omega}_d$. τ_{FF} would produce the motion you expect to need and $-K_d \tilde{\omega}$ corrects tracking errors.

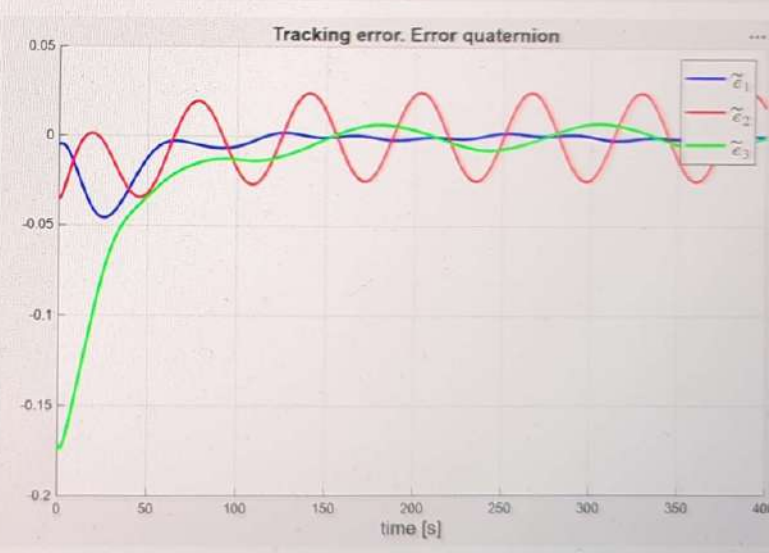
1.7

1.6


Euler angles
angle [deg]
time [s]
ϕ
θ
ψ


Control input
input [Nm]
time [s]


Angular velocities
angular rate [deg/s]
time [s]
p
q
r


Tracking error. Error quaternion
time [s]

1.7 a)

$$V = \frac{1}{2}\tilde{\omega}^T I_g \tilde{\omega} + 2k_p(1-\tilde{\eta})$$

$$\dot{V} = \frac{1}{2}\dot{\tilde{\omega}}^T I_g \tilde{\omega} + \frac{1}{2}\tilde{\omega}^T I_g \dot{\tilde{\omega}} - 2k_p\dot{\tilde{\eta}}$$

$$I_g = I_g^T$$

$$\dot{V} = \tilde{\omega}^T I_g \dot{\tilde{\omega}} - 2k_p\dot{\tilde{\eta}}$$

$$\dot{\tilde{\eta}} = -\frac{1}{2}\tilde{\varepsilon}^T\tilde{\omega}$$

$$\Rightarrow \dot{V} = \tilde{\omega}^T I_g \dot{\tilde{\omega}} + k_p\tilde{\varepsilon}^T\tilde{\omega}$$

$$\omega_d = 0 \Rightarrow \tilde{\omega} = \omega, \quad \dot{\tilde{\omega}} = \dot{\omega}$$

$$\dot{V} = \omega^T I_g \dot{\omega} + k_p\tilde{\varepsilon}^T\omega$$

$$I_g\dot{\omega} - S(I_g\omega)\omega = \tau$$

$$I_g\dot{\omega} = S(I_g\omega)\omega + \tau$$

$$\Rightarrow \dot{V} = \omega^T S(I_g\omega)\omega + \omega^T\tau + k_p\tilde{\varepsilon}^T\omega$$

$$\omega^T S(I_g\omega)\omega = 0$$

$$\dot{V} = \omega^T\tau + k_p\tilde{\varepsilon}^T\omega$$

$$\tau = -k_d\omega - k_p\tilde{\varepsilon}$$

$$\dot{V} = -\omega^T k_d\omega - k_p\omega^T\tilde{\varepsilon} + k_p\tilde{\varepsilon}^T\omega$$

$$\omega^T\tilde{\varepsilon} = \tilde{\varepsilon}^T\omega$$

$$\dot{V} = -\omega^T k_d\omega$$

$$K_d = k_d I_3$$

$$\dot{V} = -k_d\omega^T\omega$$

b) $V = \frac{1}{2}\tilde{\omega}^T I_g \tilde{\omega} + 2k_p(1-\tilde{\eta})$

$V > 0 \quad \forall (\tilde{\varepsilon}, \tilde{\omega}) \neq 0, \quad V(0) = 0$

From a) $\dot{V} = -k_d \omega^T \omega \leq 0$

$\dot{V} = 0 \iff \omega = 0$

$I_g \dot{\omega} - S(I_g \omega)\omega = \tau$
with $\omega = 0$
$0 = \tau = -k_p \tilde{\varepsilon}$

$\tilde{\varepsilon} = 0$
$\tilde{\eta} = 1$

Krasovskii-LaSalle : 0,0 is A.S

c) No,
For a unit quaternion
$\tilde{\eta}^2 + \tilde{\varepsilon}^T \tilde{\varepsilon} = 1$

$\tilde{\varepsilon} = 0$
$\tilde{\eta} = \pm 1$

$(\tilde{\eta}, \tilde{\varepsilon}, \omega) = (1, 0, 0)$ and $(-1, 0, 0)$
are equilibria, since $\tau = -k_d \omega - k_p \tilde{\varepsilon} = 0$

An initial condition at $(-1, 0, 0)$
will not converge to $(1, 0, 0)$.
$\Rightarrow (1, 0, 0)$ cannot be G.A.S

G.A.S cannot be proven because
there are two quaternion equilibria